Numerical evaluation and matrix representation of the decay feed-down kernel

Notes accompanying `DecaySpectra.jl`

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These notes accompany `DecaySpectra.jl`. For the two-, three-, and four-body decay maps of the main notes (equation numbers below refer to *Maximum_Entropy_notes.pdf*) they document how each map is (i) reduced to a form that fixed-order Gauss–Legendre quadrature evaluates rapidly, and (ii) precomputed as a matrix acting on a parent spectrum sampled on a fixed grid.

How to read this. The presentation is *topic by topic*, and within each topic the *analytic* structure is kept separate from its *numerical* treatment: first what an integral *is* — its domain, its singularities, any closed-form structure — then how it is *computed*. The topics build on one another:

- the two-body kernel and its quadrature (Sec. 2);
- the effective-mass average that carries three- and four-body decays (Sec. 3);
- finite resonance widths (Sec. 4);
- the matrix representation for boost-invariant sources (Sec. 5);
- its block-Toeplitz generalization to rapidity-dependent sources (Sec. 6).

One device recurs throughout and is worth keeping in mind: every multi-body or finite-width complication is an *average of the same two-body kernel* over an effective-mass or spectral distribution. The expensive kernel is therefore evaluated once; only the cheap averaging weight changes.

1 The decay map is linear

After the analytic azimuthal integration, the daughter spectrum of the two-body decay $a \to b+c$ is [Eq. (32)]

$$\mathcal{F}_b(p_T,\eta_p,\phi_p) = \frac{B\,m_a}{4\pi\,p^a_{b|c}} \int_0^\infty q_T\,\mathrm{d}q_T \int_{-\infty}^\infty \mathrm{d}\eta_q\, \frac{\Theta\big(Q^2-D^2\big)}{\sqrt{Q^2-D^2}} \Big[\mathcal{F}_a(q_T,\eta_q,\phi_p+\alpha)+\mathcal{F}_a(q_T,\eta_q,\phi_p-\alpha)\Big], \tag{1}$$

with $M = m_{aT}m_{bT}$, $Q = q_T p_T$, $C = m_a E^a_{b|c}$, $D = M\cosh(\eta_q-\eta_p)-C$, $Z = D/Q$, $\alpha = \arccos Z$. The three-body decay [Eq. (59)] is the same kernel averaged over the effective mass μ of the $c+d$ pair with weight $p^a_b(\mu)\,p^{\tilde{c}}_c(\mu)/\mathcal{N}_{a\to bcd}$ [Eqs. (45)–(48)]. In Fourier modes, $\mathcal{F} = \sum_n \mathcal{F}_n e^{in\phi}$ [Eq. (62)], the two azimuthal roots produce the Chebyshev factor $2T_n(Z)$ [Eqs. (40), (41), (61)].

The essential observation is that (1) is *linear* in $\mathcal{F}_a$. Once the parent spectrum is represented by its values on a fixed grid, the map is a matrix, built once per decay and per mode; applying it to any source is a matrix–vector product.

2 The two-body kernel

Both decays reduce to the double integral

$$K(E_b) = \int q_T \,\mathrm{d}q_T \int \mathrm{d}\eta_q \,\frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\,\big[\text{bracket}\big], \qquad p_b = \sqrt{E_b^2 - m_b^2}, \quad C = m_a E_b, \tag{2}$$

where the bracket is the source sum of (1) (or $2T_n(Z)\mathcal{F}_{a,n}$ in mode form). The integrand lives on a compact region with inverse-square-root boundary singularities. This section has two halves. The analytic half (Sec. 2.1) fixes that region in closed form and locates every non-smooth feature: the $1/\sqrt{\ }$ window edges and a single logarithmic singularity where the domain changes topology. The numerical half (Sec. 2.2) removes each by one substitution, so that plain tensor-product Gauss–Legendre quadrature converges rapidly, and reports the measured accuracy.

2.1 Analytic structure: domain and singularities

2.1.1 The $\Delta\eta$ window

What the window is. Before the algebra, the physical content. The analytic azimuthal integration of the main notes rests on the on-shell δ-function $\delta(q\cdot p + m_a E_b)$ [Eq. (4)], which at fixed rapidities fixes the relative azimuth through $\cos\Delta\phi = Z$ [Eq. (19)], with

$$Z = \frac{M\cosh\Delta\eta - C}{Q} = \frac{D}{Q}. \tag{3}$$

A real opening angle $\Delta\phi$ exists *iff* $|Z| \le 1$ [Eq. (21)] — and $\Theta(Q^2 - D^2) = \Theta(1 - Z^2)$ is exactly the indicator of that condition. So the "$\Delta\eta$ window" answers a kinematic question: *for a daughter observed at (p_T, η_p), which parent momenta (q_T, η_q) can produce it?* The parent must sit at a rapidity separation $\Delta\eta$ for which some azimuthal orientation places the daughter at p_T; too small or too large a $\cosh\Delta\eta$ makes $|Z| > 1$ and no real $\Delta\phi$ exists. The endpoints $Z = \pm 1$ are the grazing configurations $\Delta\phi = 0, \pi$ (daughter emitted parallel or antiparallel to the parent's transverse momentum), and the $1/\sqrt{Q^2 - D^2} = 1/(Q\sqrt{1 - Z^2})$ factor is the Jacobian $|\mathrm{d}\cos\Delta\phi/\mathrm{d}\Delta\phi|^{-1}$ that diverges integrably there, as at any turning point.

The bounds. Throughout, $M = m_{aT} m_{bT} > 0$, $Q = q_T p_T \ge 0$ (with $Q > 0$ for interior nodes, $q_T, p_T > 0$), and $C = m_a E_b > 0$; all three depend on q_T and/or p_T but not on $\Delta\eta$. At fixed q_T the integrand is supported where the argument of the Θ function is positive:

$$Q^2 - D^2 \ge 0 \iff |D| \le Q \iff -Q \le M\cosh\Delta\eta - C \le Q. \tag{4}$$

Adding C and dividing by $M > 0$ (both operations preserve the inequalities) gives the two-sided condition

$$\frac{C - Q}{M} \le \cosh\Delta\eta \le \frac{C + Q}{M}. \tag{5}$$

This must be intersected with the range of the hyperbolic cosine, $\cosh\Delta\eta \geq 1$. Two of the three bounds compare trivially: $(C-Q)/M \leq (C+Q)/M$ always (they differ by $2Q/M \geq 0$). Therefore

$$\cosh\Delta\eta \;\in\; \Big[\max\Big(1, \tfrac{C-Q}{M}\Big),\; \tfrac{C+Q}{M}\Big], \tag{6}$$

and this interval is nonempty *iff* its upper end clears the lower: since $\max(1,(C-Q)/M) \leq (C+Q)/M$ reduces — whichever branch the max takes — to the single condition

$$\frac{C+Q}{M} \;\geq\; 1 \;\iff\; M \;\leq\; C+Q, \tag{7}$$

because the branch $(C-Q)/M \leq (C+Q)/M$ holds unconditionally. Condition (7) is the origin of the q_T band of Sec. 2.1.2.

To translate (6) into a set in $\Delta\eta$: on $\Delta\eta \geq 0$ the map $\Delta\eta \mapsto \cosh\Delta\eta$ is strictly increasing and continuous, so the preimage of the interval (6) is the interval $[\eta_{\rm lo}, \eta_{\rm hi}]$ with

$$\eta_{\rm hi} = \operatorname{arccosh}\frac{C+Q}{M}, \qquad \eta_{\rm lo} = \begin{cases} \operatorname{arccosh}\dfrac{C-Q}{M}, & C-Q > M, \\ 0, & C-Q \leq M, \end{cases} \tag{8}$$

(the second case includes $C-Q \leq 0$, where the lower constraint in (5) is vacuous since $\cosh \geq 1 > (C-Q)/M$). Since cosh is even, the full solution set is symmetric,

$$\big\{\Delta\eta :\; |\Delta\eta| \in [\eta_{\rm lo}, \eta_{\rm hi}]\big\} = \big[-\eta_{\rm hi}, -\eta_{\rm lo}\big] \cup \big[\eta_{\rm lo}, \eta_{\rm hi}\big] : \tag{9}$$

two disjoint intervals when $\eta_{\rm lo} > 0$ (the parent *must* be at different rapidity than the daughter), merging into the single interval $[-\eta_{\rm hi}, \eta_{\rm hi}]$ when $\eta_{\rm lo} = 0$. The change between the two topologies as q_T varies is the origin of the logarithmic singularity of Sec. 2.1.3.

Geometrically the two regimes read as follows. When $\eta_{\rm lo} > 0$ (i.e. $C-Q > M$, meaning C is large: the parent is slow or the daughter hard) the parent cannot sit at the same rapidity as the daughter — a co-moving parent would have to emit the daughter with $|Z| > 1$ — so the allowed set is two rapidity bands straddling η_p, symmetric about it. As q_T increases, Q grows, $C-Q$ eventually drops below M, the inner gap closes ($\eta_{\rm lo} \to 0$), and the bands merge: now a parent *at* the daughter's rapidity can feed it (with $\Delta\phi$ given by the interior value $Z(0) = (M-C)/Q \in (-1,1)$). The merging point is exactly q_T^* of Sec. 2.1.3. The upper edge $\eta_{\rm hi}$ always exists (some maximal rapidity separation beyond which the boost carries the daughter past p_T for every azimuth) and is always a genuine $1/\sqrt{\ }$ singularity; the lower edge is singular only in the two-band regime. Both rapidity signs $\Delta\eta \gtrless 0$ contribute; they are evaluated separately because the source need not be symmetric in rapidity. At the window edges $Z = \pm 1$ and $Q^2 - D^2$ vanishes *linearly* in $\Delta\eta$: writing $Q^2 - D^2 = (Q-D)(Q+D)$ (the factored form is also what is computed — it avoids the cancellation of subtracting two large squares), one factor vanishes at each edge while $\mathrm{d}D/\mathrm{d}\Delta\eta = M\sinh\Delta\eta \neq 0$ there, so the zero is simple and the integrand diverges as an integrable $1/\sqrt{\ }$. When $\eta_{\rm lo} = 0$ (merged window) the lower edge is *not* singular — $Z(0) = (M-C)/Q$ lies inside $(-1,1)$ — and the cosine map's clustering there is merely harmless. Two numerical guards accompany the window: q_T nodes with $(C+Q)/M \leq 1$ are skipped (roundoff can push a node marginally outside the support), and inside the loop $S_2 = (Q-D)(Q+D) \leq 0$ nodes are dropped for the same reason.

2.1.2 The q_T band

By (7), the window is nonempty exactly when

$$m_{aT}\, m_{bT} \;\le\; C + q_T p_T. \tag{10}$$

Both sides of (10) are strictly positive ($C = m_a E_b > 0$ and $m_{aT} m_{bT} > 0$), so squaring preserves the inequality in both directions and is an equivalence. Squaring and expanding with $m_{aT}^2 = m_a^2 + q_T^2$, $m_{bT}^2 = m_b^2 + p_T^2$:

$$\begin{aligned} \big(m_a^2 + q_T^2\big)\big(m_b^2 + p_T^2\big) \;&\le\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2, \\ m_a^2 m_b^2 + m_a^2 p_T^2 + m_b^2 q_T^2 + q_T^2 p_T^2 \;&\le\; C^2 + 2C p_T\, q_T + q_T^2 p_T^2. \end{aligned} \tag{11}$$

The $q_T^2 p_T^2$ terms cancel; using $m_a^2 m_b^2 + m_a^2 p_T^2 = m_a^2 m_{bT}^2$ and collecting powers of q_T,

$$m_b^2\, q_T^2 - 2C p_T\, q_T + m_a^2 m_{bT}^2 - C^2 \;\le\; 0, \tag{12}$$

a upward parabola in q_T: the inequality holds between its roots. The quarter-discriminant is

$$\begin{aligned} \tfrac{1}{4}\Delta &= (C p_T)^2 - m_b^2\big(m_a^2 m_{bT}^2 - C^2\big) = C^2\big(p_T^2 + m_b^2\big) - m_a^2 m_b^2\, m_{bT}^2 \\ &= m_{bT}^2\big(C^2 - m_a^2 m_b^2\big) = m_{bT}^2\, m_a^2\big(E_b^2 - m_b^2\big) = m_{bT}^2\, m_a^2\, p_b^2 \;\ge\; 0, \end{aligned} \tag{13}$$

where the second line used $p_T^2 + m_b^2 = m_{bT}^2$, then $C = m_a E_b$, then $p_b^2 = E_b^2 - m_b^2$. The discriminant is a perfect square — the roots are always real, as they must be (the band exists for every open decay). The quadratic formula then gives

$$q_T^{\pm} = \frac{C p_T \pm m_a p_b\, m_{bT}}{m_b^2} = \frac{m_a\big(E_b\, p_T \pm p_b\, m_{bT}\big)}{m_b^2}, \qquad q_T \in \big[\max(0, q_T^-),\, q_T^+\big]. \tag{14}$$

The sign of the lower root follows from $\big(E_b p_T\big)^2 - \big(p_b m_{bT}\big)^2 = E_b^2 p_T^2 - p_b^2\big(m_b^2 + p_T^2\big) = p_T^2\big(E_b^2 - p_b^2\big) - p_b^2 m_b^2 = m_b^2\big(p_T^2 - p_b^2\big)$, so $q_T^- > 0 \iff p_T > p_b$. The lower root is positive iff $p_T > p_b$: a daughter harder than the rest-frame momentum can only come from a moving parent, while for $p_T < p_b$ even a parent at rest contributes and the band starts at $q_T = 0$.

The band width, $q_T^+ - q_T^- = 2 m_a p_b m_{bT}/m_b^2$, carries the m_b^{-2} of the roots: light daughters have enormous bands (for $\rho \to \pi\pi$ at $p_T = 0.6\,\mathrm{GeV}$ the band is $[0.5,\, 18]\,\mathrm{GeV}$), which is why the feed-down matrices of Sec. 5 are effectively dense. In practice the reach in q_T is limited not by the band but by the source: for a thermal-like parent spectrum $\sim e^{-(m_{aT} - m_a)/T_{\mathrm{eff}}}$ the integrand is negligible beyond

$$q_T^{\max} \;=\; \sqrt{\big(m_a + T_{\mathrm{eff}} \ln(1/\varepsilon)\big)^2 - m_a^2}, \tag{15}$$

(ε the tolerated relative tail, e.g. 10^{-8}), which also sets the required extent of any fixed parent grid. Note that T_{eff} is the slope of the *actual* source — radial flow blueshifts it well above the freeze-out temperature.

2.1.3 The logarithmic singularity at q_T^*

Inside the band the window changes topology where η_{lo} reaches zero, i.e. where $C - Q - M$ changes sign (case boundary of (8)). Solving $m_{aT} m_{bT} = C - q_T p_T$ by the same squaring as in

Sec. 2.1.2 — legitimate on the branch $C - q_T p_T \geq 0$, which holds there since $m_{aT} m_{bT} > 0$ — flips only the sign of the linear term in (12), so the roots are $-q_T^{\mp}$, and the positive one is

$$q_T^* = \frac{m_a\big(p_b\, m_{bT} - E_b\, p_T\big)}{m_b^2} = -\, q_T^-, \tag{16}$$

which lies inside the band $(0, q_T^+)$ precisely when $q_T^- < 0$, i.e. $p_T < p_b$.

For the local behavior define $g \equiv Q + M - C$, so $g < 0$ is the two-interval regime ($\eta_{\rm lo} > 0$) and $g > 0$ the merged one, with $g(q_T^*) = 0$; g is strictly increasing in q_T ($\mathrm{d}g/\mathrm{d}q_T = p_T + q_T\, m_{bT}/m_{aT} > 0$), so near the transition $g \simeq g'(q_T^*)\,(q_T - q_T^*)$. Expand the integrand factors around $\Delta\eta = 0$ using $\cosh\Delta\eta = 1 + \frac{1}{2}\Delta\eta^2 + \mathcal{O}(\Delta\eta^4)$:

$$Q + D = Q + M\cosh\Delta\eta - C \;=\; g + \tfrac{1}{2} M \Delta\eta^2 + \mathcal{O}(\Delta\eta^4), \qquad Q - D \;=\; 2Q - g - \tfrac{1}{2} M\Delta\eta^2 \;\simeq\; 2Q, \tag{17}$$

the second factor staying regular (it vanishes at the *other* edge $\eta_{\rm hi}$, far from $\Delta\eta = 0$). Hence, up to a cutoff Λ separating the local region from the rest of the window,

$$\int \frac{\mathrm{d}\Delta\eta}{\sqrt{(Q-D)(Q+D)}} \;\simeq\; \frac{1}{\sqrt{2Q}} \int \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \frac{1}{2} M \Delta\eta^2}}, \tag{18}$$

and the local integral has elementary antiderivatives on both sides of the transition:

$$g > 0: \quad \int_0^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{g + \frac{1}{2} M\Delta\eta^2}} = \sqrt{\tfrac{2}{M}}\, \operatorname{arcsinh}\Big(\Lambda\sqrt{\tfrac{M}{2g}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln g\Big] + \mathcal{O}(g), \tag{19}$$

$$g < 0: \quad \int_{\eta_{\rm lo}}^{\Lambda} \frac{\mathrm{d}\Delta\eta}{\sqrt{\frac{1}{2} M\big(\Delta\eta^2 - \eta_{\rm lo}^2\big)}} = \sqrt{\tfrac{2}{M}}\, \operatorname{arccosh}\Big(\frac{\Lambda}{\eta_{\rm lo}}\Big) = \sqrt{\tfrac{2}{M}}\Big[\ln\big(2\Lambda\sqrt{\tfrac{M}{2}}\big) - \tfrac{1}{2}\ln|g|\Big] + \mathcal{O}(g), \tag{20}$$

where the second line used $g + \frac{1}{2} M\Delta\eta^2 = \frac{1}{2} M(\Delta\eta^2 - \eta_{\rm lo}^2)$ with $\eta_{\rm lo} = \sqrt{-2g/M}$ and $\operatorname{arccosh} x = \ln(2x) + \mathcal{O}(x^{-2})$. Both sides carry the *same* coefficient $-\frac{1}{2}\sqrt{2/M}\,\ln|g|$: the inner integral has a symmetric, integrable logarithmic singularity

$$I(q_T) \;=\; -\frac{1}{\sqrt{2QM}}\, \ln\big|q_T - q_T^*\big| \;+\; (\text{regular}), \tag{21}$$

after inserting the linearization of g. A Gauss rule straddling a logarithm converges only algebraically, so the q_T interval is split at q_T^* whenever $q_T^- < 0 < q_T^* < q_T^+$; the residual $\theta\ln\theta$ endpoint behavior after the cosine map is discussed in Sec. 2.2.1.

2.2 Numerical treatment: the cosine-mapped Gauss rule

Every non-smooth feature located above is removed by a single substitution: the $1/\sqrt{\ }$ window edges and the $\sqrt{\ }$ zeros of the μ average both cancel against a cosine map's Jacobian, leaving the lone q_T^* logarithm, which is isolated by splitting the interval there.

2.2.1 The cosine substitution and the tensor-product rule

Every one-dimensional interval $[x_{\rm lo}, x_{\rm hi}]$, of midpoint $m = \frac{1}{2}(x_{\rm lo} + x_{\rm hi})$ and half-width $h = \frac{1}{2}(x_{\rm hi} - x_{\rm lo})$, is traversed via

$$x(\theta) = m - h\cos\theta, \qquad \theta \in (0,\pi), \qquad \mathrm{d}x = h\,\sin\theta\,\mathrm{d}\theta, \tag{22}$$

which maps $\theta = 0 \mapsto x_{\rm lo}$ and $\theta = \pi \mapsto x_{\rm hi}$ monotonically. The endpoint expansion is exact in half-angle form:

$$x - x_{\rm lo} = h\,(1-\cos\theta) = 2h\sin^2\tfrac{\theta}{2}, \qquad x_{\rm hi} - x = h\,(1+\cos\theta) = 2h\cos^2\tfrac{\theta}{2}. \tag{23}$$

Consider an integrand with the inverse-square-root endpoint behavior established in Sec. 2.1.1, $f(x) = g(x)/\sqrt{(x-x_{\rm lo})(x_{\rm hi}-x)}$ with g smooth. Using (23) and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$,

$$f\big(x(\theta)\big)\,\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{g\big(x(\theta)\big)\;h\cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sqrt{2h\sin^2\frac{\theta}{2}\;\cdot\;2h\cos^2\frac{\theta}{2}}} = g\big(x(\theta)\big)\,: \tag{24}$$

the singular factors cancel *identically*, not just to leading order — the θ-integrand is as smooth as g, and Gauss–Legendre in θ converges spectrally. (If only one endpoint is singular the cancellation happens there and the other endpoint merely gains harmless node clustering; endpoint zeros $\propto \sqrt{x - x_{\rm lo}}$, as in the μ integral, become smooth $\propto \sin^2\frac{\theta}{2}$ by the same algebra.)

The θ integral itself is discretized by an n-point Gauss–Legendre rule with nodes $x_i \in (-1,1)$ and weights w_i, mapped affinely to $(0,\pi)$: $\theta_i = \frac{\pi}{2}(x_i+1)$, $\int_0^\pi G\,\mathrm{d}\theta \approx \frac{\pi}{2}\sum_i w_i G(\theta_i)$. Combining with (22),

$$\int_{x_{\rm lo}}^{x_{\rm hi}} f(x)\,\mathrm{d}x \;\approx\; h\sum_{i=1}^{n} v_i\, f\big(m - h\,c_i\big), \qquad c_i = \cos\theta_i, \quad v_i = \tfrac{\pi}{2}\,w_i\sin\theta_i, \tag{25}$$

so the map costs no trigonometry at evaluation time: the fused pairs (c_i, v_i) are computed once per order and shared (memoized) across all decay objects. Applying (25) in both directions gives the tensor-product kernel sum

$$K(E_b) \;\approx\; \sum_{\rm seg}\; h_q \sum_{i=1}^{n_q} v_i^q\; q_T^{(i)}\; h_\eta\Big(q_T^{(i)}\Big) \sum_{j=1}^{n_\theta} \frac{v_j^\theta}{\sqrt{Q^2 - D_{ij}^2}}\,\big[\text{bracket}\big]_{ij}, \tag{26}$$

where the inner $1/\sqrt{\ }$ is evaluated in the factored form $(Q-D)(Q+D)$ and the singular endpoint behavior is tamed by (24) through the mapped $\Delta\eta$ nodes. The only residual non-smoothness is the logarithm of Sec. 2.1.3 at the segment endpoint q_T^*: with $x - q_T^* \propto \theta^2$ from (23), the integrand behaves as $\sin\theta\cdot\ln\theta^2 \sim 2\,\theta\ln\theta$ near $\theta = 0$ — continuous but not analytic — for which the Gauss error decreases algebraically, $\sim n_q^{-4}$ (measured: the error at low p_T drops $\approx 16\times$ per doubling of n_q).

2.2.2 Three-body: the μ average

For $a \to b + c + d$ the kernel is averaged over the effective mass $\mu \in [m_c + m_d,\, m_a - m_b]$ with weight $p_c^{\tilde{c}}(\mu)$ and normalization $\mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\, p_c^{\tilde{c}}(\mu)$ [Eqs. (45)–(48)]. The integrand vanishes as a square root at both endpoints: $p_c^{\tilde{c}} \sim \sqrt{\mu - (m_c + m_d)}$ at threshold, and $K \propto$ band width $\propto p_b^a(\mu) \sim \sqrt{m_a - m_b - \mu}$ at the upper limit. The same substitution (22) turns $\sqrt{\ }$-type zeros into smooth θ^2 behavior, and the μ integral is done with an n_μ-point cosine-mapped Gauss rule; the nodes μ_i, energies $E_b(\mu_i)$ and normalized weights are precomputed at construction.

2.2.3 Measured convergence and default orders

For a Boltzmann source ($T = 0.15\,\mathrm{GeV}$) relative errors against high-order references behave as follows (worst case is low p_T, where the q_T^* logarithm is inside the band):

orders		error at $p_T = 0.2$	error at $p_T = 0.8$
$\rho \to \pi\pi$:	$(n_q, n_\theta) = (128, 32)$	$7 \cdot 10^{-8}$	$5 \cdot 10^{-12}$
	$(64, 16)$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-12}$
	$(32, 8)$	$3 \cdot 10^{-6}$	$2 \cdot 10^{-8}$
$\omega \to 3\pi$:	$(n_\mu, n_q, n_\theta) = (32, 128, 32)$	$2 \cdot 10^{-8}$	$3 \cdot 10^{-13}$
	$(16, 64, 16)$	$5 \cdot 10^{-7}$	$3 \cdot 10^{-10}$
	$(8, 32, 8)$	$1 \cdot 10^{-4}$	$5 \cdot 10^{-5}$

The $\Delta\eta$ direction is spectrally convergent and over-resolved already at $n_\theta = 16$; n_q is limited by the logarithm ($\sim n_q^{-4}$, only relevant for $p_T < p_b$); $n_\mu \geq 16$ is safe while $n_\mu = 4$ fails at the percent level. The defaults $(n_q, n_\theta, n_\mu) = (128, 32, 32)$ are conservative; cost scales linearly in each order. Number conservation, $N_b = B\,N_a$, holds at the 10^{-5} level for both decays at default orders.

3 Multi-body decays: the effective-mass average

The three-body μ average of Sec. 2 is the first instance of a general pattern: any number of unobserved daughters can be clustered into a single effective particle of mass μ, and the same two-body kernel is averaged over that cluster's mass. This section makes the pattern explicit — its *analytic* form is a phase-space recursion, its *numerical* cost is one extra mass quadrature per additional daughter — and applies it to four-body decays. (Finite widths, Sec. 4, are the very same construction with a spectral function in place of the phase-space weight.) The organizing question throughout is whether the extra integration multiplies the number of *expensive* kernel evaluations or only enriches the *cheap* weight — and here, as we will see, it is only the weight.

3.1 Recursive effective-mass structure

Return to the three-body average and read it structurally. For $a \xrightarrow{\sim} b + (cd)$ the observed daughter b recoils against the pair treated as a single effective particle $\widetilde{cd}$ of mass μ; the *angular* distribution of b — hence the entire $(q_T, \Delta\eta)$ kernel — depends on the pair only through μ, via $E_b(\mu) = (m_a^2 + m_b^2 - \mu^2)/2m_a$. The internal breakup of the pair enters solely through *how much probability sits at each* μ, i.e. the invariant-mass density of the pair,

$$\rho_2(\mu; m_c, m_d) \;\propto\; q(\mu; m_c, m_d), \qquad q(M; m_1, m_2) \equiv \frac{\lambda^{1/2}(M^2, m_1^2, m_2^2)}{2M}, \tag{27}$$

which is precisely the weight $p_c^{\tilde{c}}(\mu)$ of Eq. (47). Writing the map this way,

$$\mathcal{F}_b = \frac{B\,m_a}{4\pi\,\mathcal{N}} \int_{\mu_-}^{\mu_+} \mathrm{d}\mu\; \underbrace{p_b^a(\mu)\,\rho_2(\mu)}_{\text{weight}}\; K\big(E_b(\mu)\big), \qquad \mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu)\,\rho_2(\mu), \tag{28}$$

makes the generalization immediate. For an n-body decay $a \to b + (c_1 \cdots c_{n-1})$, cluster the $n-1$ unobserved particles into $\widetilde{c}$ of mass $\mu \in [\sum_i m_{c_i},\, m_a - m_b]$; the kernel is unchanged, $K(E_b(\mu))$, and only the weight generalizes to the invariant-mass density $\rho_{n-1}(\mu)$ of the cluster.

Where the density comes from. The density and its recursion are the standard factorization of n-body Lorentz-invariant phase space. Peeling the observed particle b against the cluster of invariant mass μ,

$$\mathrm{d}\Phi_n\big(m_a^2;\, p_b, \{p_{c_i}\}\big) = \frac{\mathrm{d}\mu^2}{2\pi}\, \mathrm{d}\Phi_2\big(m_a^2;\, m_b^2, \mu^2\big)\, \mathrm{d}\Phi_{n-1}\big(\mu^2;\, \{p_{c_i}\}\big), \tag{29}$$

and integrating the two-body factor over the isotropic emission angle, $\int \mathrm{d}\Phi_2(m_a^2; m_b^2, \mu^2) \propto q(m_a; m_b, \mu) = p_b^a(\mu)$, identifies the μ-measure as $p_b^a(\mu)\,\mathrm{d}\mu$ times the cluster's own phase-space volume. Defining that volume as the density, $\rho_{n-1}(\mu)\,\mathrm{d}\mu \equiv \int \mathrm{d}\Phi_{n-1}(\mu^2; \{p_{c_i}\})$ at fixed cluster mass, and applying (29) once more to $\mathrm{d}\Phi_{n-1}$ — peeling c_1 against the sub-cluster of mass σ — gives the Byckling–Kajantie recursion, one particle at a time,

$$\rho_k\big(\mu; m_{c_1}, \ldots, m_{c_k}\big) = \int_{\sigma_-}^{\sigma_+} \mathrm{d}\sigma\; q(\mu; m_{c_1}, \sigma)\; \rho_{k-1}\big(\sigma; m_{c_2}, \ldots, m_{c_k}\big), \qquad \rho_1(\sigma; m) = \delta(\sigma - m), \tag{30}$$

with $\sigma \in [\sum_{i\geq 2} m_{c_i},\, \mu - m_{c_1}]$. The overall constants (2π from (29), the 4π from each angular integration) are μ-independent and cancel in the normalized ratio $\mathcal{F}_b$, so they are dropped. The recursion terminates at $\rho_1 = \delta$ (a single particle has definite mass); one step gives $\rho_2 = q$ (the three-body weight of Eq. 47), and for four-body

$$\rho_3(\mu; m_c, m_d, m_e) = \int_{m_d+m_e}^{\mu-m_c} \mathrm{d}\sigma\; q(\mu; m_c, \sigma)\; q(\sigma; m_d, m_e). \tag{31}$$

The four-body map explicitly. Assembling the pieces, the observed-b spectrum of $a \to b\,c\,d\,e$ is the two-dimensional mass average

$$\mathcal{F}_b = \frac{B\, m_a}{4\pi\, \mathcal{N}} \int_{m_c+m_d+m_e}^{m_a-m_b} \mathrm{d}\mu\; p_b^a(\mu) \underbrace{\left[\int_{m_d+m_e}^{\mu-m_c} \mathrm{d}\sigma\; q(\mu; m_c, \sigma)\, q(\sigma; m_d, m_e)\right]}_{\rho_3(\mu)} K\big(E_b(\mu)\big), \quad \mathcal{N} = \int \mathrm{d}\mu\, p_b^a(\mu) \rho_3(\mu). \tag{32}$$

The bracket is the Lorentz-invariant phase-space volume of the three-particle cluster $c\,d\,e$ at total mass μ — geometrically, the area of its Dalitz plot. The inner σ integral runs over the (de) sub-mass; by the symmetry of phase space any pairing gives the same ρ_3, so the choice of which particle to peel first is free. Both mass integrals carry the $\sqrt{\ }$-type endpoint zeros of the Källén function (each q vanishes at its threshold), regularized by the cosine map of Sec. 2.2.1.

Cost. The crucial point is that $\rho_{n-1}(\mu)$ is *independent of the kernel*: it is a static, $(n-2)$-fold nested phase-space integral in the mass variables alone, computed once on the n_μ nodes at construction. The number of expensive kernel evaluations therefore stays n_μ — the *same* as three-body — for any number of unobserved daughters; only the weight becomes a nested (but kernel-free, hence cheap) integral. A four-body decay matrix costs essentially what a three-body one costs. The inner mass integrals inherit the same $\sqrt{\ }$ endpoint zeros as the μ integral (the Källén function vanishes at each threshold and pseudo-threshold), and are handled by the same cosine-mapped Gauss rule of Sec. 2.2.1. For general n the weight is an $(n-2)$-fold nested mass integral; with n_σ points per level it costs $n_\mu\, n_\sigma^{n-2}$ evaluations of the elementary function q — arithmetic only, no kernel — against the fixed n_μ kernel evaluations. So the build cost is dominated by the kernel until n is large, and a four- or five-body matrix is, in practice, a three-body matrix with a slightly richer weight.

On observing a cluster particle. In the cascade every daughter is observed in turn (Sec. 5). When the observed particle is one of the "cluster" members, one simply relabels: the observed particle plays the role of b and the remaining $n-1$ form the cluster. The construction above is symmetric under that relabelling, so nothing new is required — the observed slot is chosen, the rest are clustered, and ρ_{n-1} is built from their masses.

4 Finite widths

So far every mass has been sharp. A resonance of width Γ_i is instead spread over a spectral function — a relativistic Breit–Wigner,

$$\varrho_i(m) \;\propto\; \frac{m^2\,\Gamma_i(m)}{\big(m^2-m_i^2\big)^2+m_i^2\,\Gamma_i(m)^2}, \qquad \int \varrho_i(m)\,\mathrm{d}m = 1, \tag{33}$$

whose width is itself mass dependent when the resonance decays to two particles in relative angular momentum ℓ,

$$\Gamma_i(m) = \Gamma_i\,\frac{m_i}{m}\left(\frac{q(m;m_1,m_2)}{q(m_i;m_1,m_2)}\right)^{2\ell+1}\frac{B_\ell\big(q(m)\big)}{B_\ell\big(q(m_i)\big)} \tag{34}$$

(B_ℓ the Blatt–Weisskopf barrier factor). It vanishes at threshold $m = m_1+m_2$, so ϱ_i carries no weight below threshold, and $\varrho_i \to \delta(m-m_i)$ as $\Gamma_i \to 0$.

One principle, two costs. Whether a given width is cheap or expensive to include is decided by *how that mass enters the decay map.* A mass that enters *only through the effective cluster mass* μ — any of the unobserved daughters — reshapes the precomputed weight $\rho_{n-1}(\mu)$ and nothing else, so it costs nothing extra. A mass that enters the *kinematics of the observed leg* — the parent (through $C = m_a E_b$ and the q_T band) or the observed daughter (through E_b, p_b) — changes the kernel itself, and must be handled by an *outer average*: one kernel build per mass node. The three places a width can sit map onto exactly this split:

- **Unobserved / cluster masses — cheap.** A width on any c_i just averages the sharp momentum inside the phase-space recursion (30), $q(\mu; m_{c_1}, \sigma) \to \int \mathrm{d}m_{c_1}\,\varrho(m_{c_1})\,q(\mu;m_{c_1},\sigma)$. It modifies $\rho_{n-1}(\mu)$ and touches no kernel.
- **Observed daughter b — expensive, but usually moot.** $K \to \int \mathrm{d}m_b\,\varrho_b(m_b)\,K$. The measured π, K, p, Λ are stable or narrow, so this collapses to the sharp δ; it only matters for a broad observed resonance mid-cascade.
- **Parent a — expensive, and dominant.** $\int \mathrm{d}m_a\,\varrho_a(m_a)\,(\cdots)$. The broadest states $(\sigma, \rho, \Delta, \dots)$ are parents, so this is where finite width matters most.

Which case this cascade needs. The split above is general, but the cascade of Sec. 5 decays every resonance in *its own* mass-ordered step: $a \to bR$ and $R \to cc'$ are two separate two-body decays, never collapsed into one three-body $a \to bcc'$. A broad R therefore enters *as a parent* — when R itself decays — not as a cluster mass. The cluster case is the *collapsed* picture, which this cascade does not use; and the genuine multi-body channels it does treat by the effective-mass integral ($\omega \to 3\pi$, $\eta \to 3\pi$) have *stable* daughters, carrying no cluster width either. So

for this cascade the parent average — the *expensive* case — is the main use of the per-species widths in the decay list. The cheap case is kept for completeness (a collapsed treatment, or a genuinely broad unobserved daughter), but it is the exception here. The next two subsections treat each in turn, cheap first.

4.1 The cheap case: width is a reweighting of the μ integral

Case 1 is not a new mechanism — it *is* the effective-mass integral of Sec. 3.1. When the unobserved pair $c\,c'$ forms a resonance R, its density $\rho_2(\mu) = q(\mu; m_c, m_{c'})$ picks up the R propagator and *becomes* a Breit–Wigner peak in μ. The two statements

$$\text{“}a \to b\,R\text{, then } R \to c\,c' \text{ with } R \text{ broad”} \quad \Longleftrightarrow \quad \text{“}a \to b\,c\,c'\text{, pair weighted by } \varrho_R(\mu)\text{”}$$

are literally the same integral (28). A narrow R collapses $\varrho_R \to \delta(\mu - m_R)$ and factorizes it back into the sharp two-body steps $a \to bR$ then $R \to cc'$; a broad R smears the two together. This is the elegant limit — the width lives entirely in a weight the effective-mass integral already computes, a change to the density rather than the kernel. But it is exactly the *collapsed* step the sequential cascade avoids: there the same broad R reappears as a parent in its own decay (next subsection), which is where its width acts. So this route, though cheap, is the exception in this cascade, not the rule.

4.2 The main case: parent width as a matrix average

For the parent (or observed daughter) the mass reshapes the kinematics, so there is no weight to hide it in. But the matrix form (Sec. 5) keeps it clean: the sharp-mass map is linear in the source and m_a enters only kinematically, so the width average is a weighted sum of sharp matrices,

$$W_n^{\text{eff}} = \int \mathrm{d}m_a\, \varrho_a(m_a)\, W_n(m_a) \;\approx\; \sum_i w_i\, W_n\big(m_a^{(i)}\big), \tag{35}$$

built at n_{m_a} mass nodes across the support of ϱ_a. The result is a matrix of the *same shape* — cascade, apply and persistence are untouched, and the width is absorbed once, at build time. (Interpolating W_n in m_a instead of rebuilding fails: the q_T band $[q_T^-, q_T^+]$ of (14) *moves* with m_a, so the matrices have shifting support and do not interpolate.) Two consequences follow.

A width opens closed channels. At the sharp pole mass a channel with $\sum_{\text{daughters}} m > m_a$ is forbidden, and the cascade skips it. But ϱ_a has support above the pole: on the tail $m'_a > \sum m$ the daughters fit, and the channel contributes. Nothing special is required — the same mass average (35) does it, with the closed nodes ($m'_a < \sum m$) simply contributing zero:

$$W_n^{\text{eff}}\big|_{\text{closed channel}} = \int_{\sum m}^{m_{\max}} \mathrm{d}m'_a\; \varrho_a(m'_a)\, W_n(m'_a). \tag{36}$$

A channel that is exactly zero in the sharp limit thus acquires a small weight from the parent's high-mass tail — this is the physical meaning of the “closed-at-nominal-mass” count the cascade reports.

Production and decay use the same ϱ_a. In a thermal source the parent is produced already dressed with its spectral function, so $\mathcal{F}_a$ carries ϱ_a; the decay then needs the parent's mass *conditional* on its momentum. The standard approximation uses the same ϱ_a in both places — exact as $\Gamma_a \to 0$, and controlled otherwise. The factorized form here ($\mathcal{F}_a$ times the ϱ_a-averaged decay) is the usual on-shell-spectral compromise, good while Γ_a/m_a is not large.

Status. Four-body is implemented (the 531 cluster keys of Sec. 3.1), and so now is the *parent width* — the main case for this cascade. Each species with $\Gamma > 1\,\mathrm{MeV}$ is averaged over its (constant-width, truncated) Breit–Wigner by (35): the decay matrix at n_{m_a} parent-mass nodes, weighted by ϱ_a and summed, giving a same-shape matrix that the cascade uses unchanged. The mass integral uses the tan-substitution $s = m_a^2 + m_a m_R \Gamma \tan\psi$, which flattens the Lorentzian so plain Gauss–Legendre in ψ resolves the peak for any Γ (a cosine map would starve it). Closed nodes ($m_a < \sum m_{\mathrm{daughters}}$) enter only the normalization, so a channel closed at the pole keeps its open-tail fraction — channel reopening, for free. Validated by number conservation, the single-node $\to$ sharp and narrow-Γ $\to$sharp limits, spectral convergence in n_{m_a}, and reopening; on the full list ~ 3200 of the 18,000 keys are broad, shifting the feed-down at the few-percent level. Since the averaged matrix is the *same shape*, the cache does not grow — only the build (an n_{m_a}-fold repeat for broad-parent keys), so persistence (Sec. 5) is essential. The cheap cluster reweight (for a genuine broad unobserved daughter) and observed-daughter width (rare) remain future work.

5 The matrix representation: one matrix per mode

Let the parent mode $\mathcal{F}_{a,n}$ be independent of η_q and sampled on a grid $q_T^{(1)}, \ldots, q_T^{(N_q)}$, and let interpolation reconstruct it anywhere via local Lagrange cardinal weights L_j of order P (stencil of P points, error $\mathcal{O}(h^P)$),

$$\mathcal{F}_{a,n}(q_T) \;\approx\; \sum_j L_j(q_T)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \qquad L_j\big(q_T^{(i)}\big) = \delta_{ij}. \tag{37}$$

First the mode bookkeeping. Insert the mode input $\mathcal{F}_a(q_T, \eta_q, \phi_q) = \mathcal{F}_{a,n}(q_T)\cos(n\phi_q)$ into the source bracket of (1), evaluated at $\phi_p = 0$; together with the two rapidity signs (equal contributions, since neither the kernel nor this source distinguishes $\pm\Delta\eta$) the four source evaluations are

$$\sum_{s=\pm}\sum_{r=\pm} \mathcal{F}_{a,n}(q_T)\cos\big(n(0{+}r\alpha)\big) = \mathcal{F}_{a,n}(q_T)\left[2\cos(n\alpha){+}2\cos(-n\alpha)\right] = 4\cos(n\alpha)\,\mathcal{F}_{a,n}(q_T) = 4\,T_n(Z)\,\mathcal{F}_{a,n}(q_T), \tag{38}$$

using the evenness of the cosine and $\cos(n \arccos Z) = T_n(Z)$ [Eq. (40)]. Now discretize (1) by the tensor rule (26) and insert the interpolation (37) at every q_T node:

$$\mathcal{F}_{b,n}\big(p_T^{(k)}\big) \approx \frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta}{\sqrt{Q^2 - D_{ij'}^2}}\, 4T_n(Z_{ij'}) \underbrace{\sum_j L_j\big(q_T^{(i)}\big)\, \mathcal{F}_{a,n}\big(q_T^{(j)}\big)}_{\text{interpolated source}}$$

$$= \sum_{j=1}^{N_q} \underbrace{\left[\frac{B\, m_a}{4\pi p_b} \sum_{\text{seg}} h_q \sum_i v_i^q\, q_T^{(i)}\, h_\eta^{(i)} \sum_{j'} \frac{v_{j'}^\theta\, 4T_n(Z_{ij'})}{\sqrt{Q^2 - D_{ij'}^2}}\, L_j\big(q_T^{(i)}\big) \right]}_{= W_n[k,j]} \mathcal{F}_{a,n}\big(q_T^{(j)}\big), \tag{39}$$

where the second line only exchanged the finite sums — legitimate because every factor except $\mathcal{F}_{a,n}$ is independent of the source. The matrix element $W_n[k,j]$ is thus the exact quadrature of the kernel against the j-th cardinal function. Assembly therefore runs the *same* quadrature loops as a functor evaluation; wherever the source would be evaluated, the bracketed multiplier is instead scattered onto the P columns of the interpolation stencil of $q_T^{(i)}$ ("scatter assembly"). One row costs one kernel evaluation with no source calls; the whole matrix costs N_{p_T} of them. Equivalently — and this was used as a cross-check — column j of W_n equals the functor applied to the j-th cardinal basis function, by the same linearity; the scatter form merely avoids the N_q-fold redundancy of computing columns one at a time.

Two properties of the quadrature matter for the matrix:

- The Θ-support in q_T is the band $q_T^\pm = m_a\big(E_{b|c}^a\, p_T \pm p_{b|c}^a\, m_{bT}\big)/m_b^2$; each row is supported (up to the stencil halo) inside its band. For light daughters the band is wide ($\propto m_b^{-2}$), so in practice the matrices are effectively dense (measured fill 65–76% for $\rho \to \pi\pi$, $\omega \to \pi\pi\pi$), and they are numerically full rank; neither sparse nor low-rank storage pays. At $N_{p_T} \times N_q \sim 39 \times 121$ a matrix is $\sim 37\,\text{KB}$ and applies in microseconds.

- The integrand has an integrable $\ln|q_T - q_T^*|$ singularity at $q_T^* = -q_T^-$, where the $\Delta\eta$ domain splits; the interval is divided there. All $1/\sqrt{\ }$ endpoint singularities (in $\Delta\eta$, at $Z = \pm 1$) and $\sqrt{\ }$-type endpoint zeros (in μ) are absorbed by the substitution $x = \text{mid} - \text{half}\cos\theta$, whose $\sin\theta$ Jacobian regularizes them; fixed-order Gauss–Legendre then converges spectrally except for the residual $\theta \ln\theta$ behavior at q_T^* ($\sim n_q^{-4}$).

The only error introduced relative to the functor is interpolation, $\mathcal{O}(h^P)$; with a cubic stencil ($P = 4$) on $h = 0.035\,\text{GeV}$ a thermal source reproduces the functor to a few $\times 10^{-5}$, and the truncation "$\mathcal{F}_a \equiv 0$ outside the grid" requires the source to be negligible beyond the last grid point.

5.1 The Z-space form of the assembly integral

For the matrix assembly the inner rapidity integral can be freed of all transcendental functions. The observation is that an η-independent source never requires the rapidity $\Delta\eta$ itself — only $Z = D/Q$ enters, through the Chebyshev factor $T_n(Z)$.

5.1.1 Derivation

Start from the inner integral of the mode-n assembly at fixed q_T, with both rapidity signs folded in (factor 2, legitimate because the kernel is even in $\Delta\eta$ and the source does not depend on it) and both azimuthal roots summed (the $2T_n$ of Eq. 41):

$$I_n \;=\; 2\int_{\eta_{\rm lo}}^{\eta_{\rm hi}} \frac{\mathrm{d}\Delta\eta}{\sqrt{Q^2-D^2}}\, 2\,T_n(Z), \qquad D = M\cosh\Delta\eta - C, \quad Z = \frac{D}{Q}. \tag{40}$$

Substitute $c = \cosh\Delta\eta$, $\mathrm{d}\Delta\eta = \mathrm{d}c/\sqrt{c^2-1}$, which maps the window (8) to $c \in [\,\max(1, (C-Q)/M),\, (C+Q)/M\,]$; then pass to

$$Z = \frac{Mc - C}{Q}, \qquad c(Z) = \frac{C + QZ}{M}, \qquad \mathrm{d}c = \frac{Q}{M}\,\mathrm{d}Z, \qquad \sqrt{Q^2-D^2} = Q\sqrt{1-Z^2}. \tag{41}$$

The window endpoints become $Z = 1$ (kinematic edge, $c = (C+Q)/M$) and $Z = Z_{\rm lo}$ with

$$Z_{\rm lo} = \max\!\left(-1,\; \frac{M-C}{Q}\right), \tag{42}$$

the value $(M-C)/Q$ corresponding to $c = 1$, i.e. $\Delta\eta = 0$. Collecting the Jacobians,

$$I_n = \frac{4}{M}\int_{Z_{\rm lo}}^{1} \frac{T_n(Z)\,\mathrm{d}Z}{\sqrt{(1-Z^2)\,(c(Z)^2-1)}}. \tag{43}$$

This is an exact change of variables, not an approximation: the $\Delta\eta$ form of Sec. 2 and (43) are the same number. What has changed is the computational content of one quadrature node: $c(Z)$ is a multiply–add, $T_n(Z)$ follows from the recurrence $T_{k+1} = 2ZT_k - T_{k-1}$ [Eq. (40)], and the only remaining special function is one square root. No arccosh (window bounds), cosh (per node) or arccos (azimuthal angle) appears.

5.1.2 Quadrature: two regimes

Full window, $Z_{\rm lo} = -1$ (i.e. $C - Q \geq M$). The factor $1/\sqrt{1-Z^2}$ is exactly the Chebyshev–Gauss weight, so with the standard nodes the weight is integrated *exactly*:

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta} \frac{T_n(Z_k)}{\sqrt{c(Z_k)^2-1}}, \qquad Z_k = \cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}. \tag{44}$$

The nodes are precomputed once per order; note also $T_n(Z_k) = \cos(n\theta_k)$, so for fixed n even the Chebyshev factors are constants of the rule.

Partial window, $-1 < Z_{\rm lo} < 1$ (i.e. $C - Q < M$). Both endpoint factors vanish like square roots *inside* the domain: $1 - Z^2$ at $Z = 1$, and $c^2 - 1$ at $Z = Z_{\rm lo}$. The decisive point is that $c - 1 = \frac{Q}{M}(Z - Z_{\rm lo})$ is *exactly linear* in Z (immediate from $c(Z) = (C+QZ)/M$ and $C + QZ_{\rm lo} = M$), so the full radicand splits into an endpoint-singular part and a manifestly regular remainder,

$$\big(1-Z^2\big)\big(c^2-1\big) = \underbrace{(1-Z)(Z-Z_{\rm lo})}_{\text{endpoint zeros}}\;\underbrace{\tfrac{Q}{M}\,(1+Z)\,(c+1)}_{\equiv\; R(Z)^2}, \tag{45}$$

with $R(Z)^2$ *strictly positive on the closed interval* ($Q, M > 0$; $1 + Z > 0$ because $Z \geq Z_{\rm lo} > -1$; $c + 1 \geq 2$). The integrand is therefore exactly the form of Sec. 2.2.1, $f(Z) = g(Z)/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ with the *smooth* numerator

$$g(Z) = \frac{T_n(Z)}{R(Z)} = \frac{T_n(Z)}{\sqrt{\frac{Q}{M}(1 + Z)(c + 1)}}. \tag{46}$$

So g carries the *shifted Chebyshev weight* $1/\sqrt{(Z - Z_{\rm lo})(1 - Z)}$ on $[Z_{\rm lo}, 1]$ — the identical structure to the full window (44), only on a shifted interval — and is integrated by the *same* rule: shifted Gauss–Chebyshev, whose nodes carry the weight exactly and whose weights are the uniform π/n_θ,

$$I_n \;\approx\; \frac{4}{M}\,\frac{\pi}{n_\theta}\sum_{k=1}^{n_\theta}\frac{T_n(Z_k)}{R(Z_k)}, \qquad Z_k = m + h\cos\theta_k, \quad \theta_k = \frac{(2k-1)\pi}{2n_\theta}, \quad c_k = \frac{C + QZ_k}{M}, \tag{47}$$

m, h the midpoint and half-width of $[Z_{\rm lo}, 1]$. Equations (44) and (47) are *one formula* $I_n = \frac{4}{M}\frac{\pi}{n_\theta}\sum_k T_n(Z_k)/R(Z_k)$ on the two windows, differing only in the interval ($[-1, 1]$ vs $[Z_{\rm lo}, 1]$) and in the regular factor left in R: $R^2 = c^2 - 1$ for the full window, $R^2 = \frac{Q}{M}(1+Z)(c+1)$ for the partial. Both remove the inverse-square-root weight *analytically*, so the summand is bounded, no positivity guard is needed, and the near-edge nodes carry their correct finite contribution.

Transition zone. As $Z_{\rm lo} \to -1^+$ the smooth g acquires a square-root branch point at $Z = -1$ (from the $1+Z$ in R) just below the lower endpoint, and the fixed-order Gauss–Chebyshev error degrades — at $Z_{\rm lo} = -0.995$, $n_\theta = 32$ gives relative error $\sim 10^{-3}$. Quadrupling the node count restores it ($\sim 10^{-12}$): the partial branch densifies for $Z_{\rm lo} < -\frac{1}{2}$, exactly as the full branch does (Eq. 44 region) on the other side of the boundary.

The algebraically equivalent alternative — forming $v_i\,T_n(Z_i)/\sqrt{(1 - Z_i^2)(c_i^2 - 1)}$ with the fused weight $v_i = \frac{\pi}{2}w_i \sin\theta_i$ of (25) carrying the $\sin\theta$ against a radicand that itself vanishes at the edge — is the same number in exact arithmetic (indeed $h\,v_i/\sqrt{(1 - Z^2)(c^2 - 1)} = \frac{\pi}{2}w_i/R(Z)$), but in floating point it is a 0/0: the radicand rounds to $\lesssim 0$ for roughly 1% of nodes (worst case $\mathcal{O}(10^{-31})$ negative), where the square root would fault or return ∞. It must then be guarded ($u > 0$ or skip), which silently forfeits those nodes. The factored form (47) needs neither the guard nor the sacrifice — $R(Z)$ is regular by construction.

5.1.3 Convergence and the transition zone

The integrand of (44), $f(Z) = T_n(Z)/\sqrt{c^2 - 1}$, has a branch point at $Z = Z_{\rm lo}$, *outside* the interval when $Z_{\rm lo} < -1$. For a function analytic on $[-1, 1]$ with nearest singularity at a real point $Z_0 < -1$, Gauss–Chebyshev error decays geometrically at the rate of the Bernstein ellipse through Z_0,

$$\varepsilon \;\sim\; \rho^{-2n_\theta}, \qquad \rho = |Z_0| + \sqrt{Z_0^2 - 1}. \tag{48}$$

Evaluating (48) with $Z_0 = Z_{\rm lo}$ and the working order $n_\theta = 8$:

$Z_{\rm lo}$	-3	-1.5	-1.2	-1.05
ρ	5.83	2.62	1.86	1.37
ρ^{-2n_θ}	$6 \cdot 10^{-13}$	$2 \cdot 10^{-7}$	$4.7 \cdot 10^{-5}$	$6.5 \cdot 10^{-3}$

The rule is excellent while the branch point stays away from the interval and degrades sharply as $Z_{\rm lo} \to -1^+$: the "transition zone". This failure mode was first seen empirically: the $n = 2$ matrix validation degraded to 2.3×10^{-3} while $n = 0$ remained at 10^{-4}. The asymmetry is instructive — for $n = 0$ the under-resolution errors are of a single sign in Z and largely average out under the subsequent q_T integration, whereas the sign-changing T_2 weight prevents the cancellation and exposes them. Per (48), quadrupling the order multiplies the exponent by four (ρ^{-8n_θ}), e.g. at $Z_{\rm lo} = -1.05$: $1.37^{-64} \approx 2 \cdot 10^{-9}$. The implementation therefore switches to $4n_\theta$ Chebyshev nodes whenever $Z_{\rm lo} > -\frac{3}{2}$, i.e. exactly where (48) predicts the base rule falls above the 10^{-7} level. The choice of threshold is a cost/accuracy compromise: a first attempt with the densification band extended to $Z_{\rm lo} > -3$ was accurate but slowed the cascade below the gain of the Z-space form itself.

5.1.4 Measured effect

On the full 1518-species cascade (8 threads, production orders n_q=32, n_θ=8, n_μ=8, 168-point grid):

assembly scheme	single 2-body build	cascade	n=2 validation
$\Delta\eta$-space (arccosh + cosh + arccos)	0.51 ms	20.2 s	$2.6 \cdot 10^{-5}$
$\Delta\eta$-space, arccos $\to T_n$ recurrence	0.41 ms	20.2 s	$2.6 \cdot 10^{-5}$
Z-space, naive	0.27 ms	16.8 s	$2.3 \cdot 10^{-3}$ (fails)
Z-space, densified transition	0.33 ms	18.3 s	$1.3 \cdot 10^{-4}$

This form applies only to the assembly (and any η-independent evaluation): the functor path evaluates an arbitrary source at $\eta_p \pm \Delta\eta$ and $\phi_p \pm \alpha$, so there the rapidity and the azimuthal angle are genuine arguments and the $\Delta\eta$-space form of Sec. 2 remains the appropriate one.

6 Rapidity-dependent sources: the $\Delta\eta$-Toeplitz structure

6.1 The kernel as a rapidity convolution

The boost-invariant result of Sec. 5 folded the $\pm\Delta\eta$ contributions together because the source carried no rapidity dependence. For a general source that fold is not available; instead the rapidity structure of the kernel must be exposed. Return to the mode-n decay map *before* any η-folding. The analytic azimuthal integration [Eq. (41)] gives, for a rapidity-dependent source,

$$\mathcal{F}_{b,n}(p_T, \eta_p) = \frac{B\, m_a}{4\pi p_b} \int_0^\infty \mathrm{d}q_T\, q_T \int_{-\infty}^\infty \mathrm{d}\Delta\eta\, \frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\, 2T_n(Z)\, \mathcal{F}_{a,n}\big(q_T, \eta_p + \Delta\eta\big), \tag{49}$$

where the parent rapidity has been written $\eta_q = \eta_p + \Delta\eta$ and $\mathrm{d}\eta_q = \mathrm{d}\Delta\eta$ at fixed η_p; every kinematic quantity $D = M\cosh\Delta\eta - C$, $Z = D/Q$ depends on rapidity only through $\Delta\eta$. The daughter rapidity η_p therefore appears in (49) *only* as the shift of the source's argument — the statement that the decay map commutes with longitudinal boosts, $\eta_p \to \eta_p + \text{const}$ carrying η_q along with it.

Collect everything that multiplies the source into a *profile*

$$K_n(p_T, q_T, \Delta\eta) \;\equiv\; \frac{B\, m_a}{4\pi p_b}\, q_T\, \frac{\Theta\big(Q^2 - D^2\big)}{\sqrt{Q^2 - D^2}}\, 2T_n(Z(q_T, \Delta\eta)), \tag{50}$$

so that (49) reads

$$\mathcal{F}_{b,n}(p_T,\eta_p)=\int_0^\infty \mathrm{d}q_T\int_{-\infty}^{\infty}\mathrm{d}\Delta\eta\; K_n(p_T,q_T,\Delta\eta)\;\mathcal{F}_{a,n}(q_T,\eta_p+\Delta\eta)\,: \tag{51}$$

at each (p_T,q_T) this is a *convolution* of the source in rapidity with the fixed profile $K_n(p_T,q_T,\cdot)$. Three properties of the profile follow directly from the kernel analysis of Sec. 2.1.1 and carry the entire structure of what follows:

1. **Even:** $K_n(\,\cdot,-\Delta\eta)=K_n(\,\cdot,\Delta\eta)$, since D, Z, and $\Theta(Q^2-D^2)$ all depend on $\Delta\eta$ through $\cosh\Delta\eta$ alone. The profile is a symmetric rapidity point-spread function; it does *not* know the sign of the boost.

2. **Compact support:** K_n vanishes outside $|\Delta\eta|\in[\eta_{\rm lo},\eta_{\rm hi}]$ (the window of Sec. 2.1.1), a set of total rapidity extent $\le 2\,\eta_{\rm hi}=2\,\mathrm{arccosh}[(C+Q)/M]$, in practice 1–2 units.

3. **η_p-independent:** K_n depends on p_T and q_T but not on where in rapidity the daughter sits. This is boost invariance, and it is what makes (51) a genuine convolution rather than a general integral operator.

Physically $K_n(p_T,q_T,\cdot)$ is the rapidity smearing induced by the decay: a parent line at a single rapidity feeding daughter momentum p_T is spread by the profile into a band of daughter rapidities of width set by $\eta_{\rm hi}$. A boost-invariant source ($\mathcal{F}_{a,n}$ independent of η_q) is the eigenfunction with the source pulling out of the convolution, $\mathcal{F}_{b,n}=\big(\int K_n\,\mathrm{d}\Delta\eta\big)\mathcal{F}_{a,n}$, recovering Sec. 5 — there the $\int K_n\,\mathrm{d}\Delta\eta$ *is* the boost-invariant matrix element.

6.2 Discretization: the block-Toeplitz matrix

Sample both spectra on a common *uniform* rapidity grid $\eta_a=\eta_0+a\,h_\eta$ (uniformity is essential — it is what turns the convolution into a Toeplitz operator). Index the daughter spectrum by $(k,a)\leftrightarrow(p_T^{(k)},\eta_a)$ and the parent by $(j,b)\leftrightarrow(q_T^{(j)},\eta_b)$. Discretize (51) in two stages. First the q_T integral and profile, by the same quadrature-plus-interpolation as Sec. 5, which produces the q_T scatter onto column index j. Second the source's rapidity argument: at a quadrature node with offset $\Delta\eta$, the value $\mathcal{F}_{a,n}(q_T^{(j)},\eta_a+\Delta\eta)$ is reconstructed from the η grid by the uniform Lagrange stencil,

$$\mathcal{F}_{a,n}\big(q_T^{(j)},\eta_a+\Delta\eta\big)\;\approx\;\sum_b L^\eta\big(\eta_a+\Delta\eta;\,\eta_b\big)\,\mathcal{F}_{a,n}\big(q_T^{(j)},\eta_b\big). \tag{52}$$

The crucial observation is that on a uniform grid the cardinal weight depends only on the *offset* of the evaluation point from the node in grid units,

$$L^\eta\big(\eta_a+\Delta\eta;\,\eta_b\big)=\ell\Big(\frac{(\eta_a+\Delta\eta)-\eta_b}{h_\eta}\Big)=\ell\Big(\frac{\Delta\eta}{h_\eta}-(b-a)\Big), \tag{53}$$

a function of $b-a$ and $\Delta\eta$ but *not* of a and b separately — translation invariance of the uniform grid, inherited from translation invariance of the profile. Assembling the coefficient of $\mathcal{F}_{a,n}(q_T^{(j)},\eta_b)$ in $\mathcal{F}_{b,n}(p_T^{(k)},\eta_a)$ therefore yields a quantity that depends on the rapidity indices only through $d\equiv b-a$:

$$W\big[(k,a),(j,b)\big]=w\big[k,\,j,\,d\big],\qquad d=b-a, \tag{54}$$

with

$$w[k,j,d] = \int \mathrm{d}\Delta\eta\; \widetilde{K}_n^{(k,j)}(\Delta\eta)\, \ell\Big(\tfrac{\Delta\eta}{h_\eta} - d\Big), \tag{55}$$

$\widetilde{K}_n^{(k,j)}$ being the profile after the q_T scatter (the quadrature sum over the q_T nodes whose stencil hits column j). Viewing W as an $N_\eta \times N_\eta$ array of $N_{p_T} \times N_q$ blocks $B_d = w[\cdot,\cdot,d]$, (54) says W is constant along block diagonals,

$$W = \begin{pmatrix} B_0 & B_1 & B_2 & \cdots \\ B_{-1} & B_0 & B_1 & \\ B_{-2} & B_{-1} & B_0 & \\ \vdots & & & \ddots \end{pmatrix}, \tag{56}$$

a *block-Toeplitz* matrix. Storing W densely repeats each block up to N_η times; storing the tensor $w[k,j,d]$ once is exact and lossless — it is the discrete profile, the point-spread function of Sec. 6.1 sampled at integer rapidity offsets.

Two further exact properties shrink the tensor, both inherited from the profile:

1. **Banded:** compact support of K_n (item 2 above) plus the finite reach of the η stencil limits the nonzero offsets to $|d| \le L$ with $L = \lceil \eta_{\mathrm{hi}}^{\max}/h_\eta \rceil + \frac{P}{2}$ (window half-width plus stencil halo), $\eta_{\mathrm{hi}}^{\max} = \max_{q_T} \operatorname{arccosh}[(C+Q)/M]$, in practice 1–2 units; so $B_d = 0$ for $|d| > L$ and only $2L+1$ offsets exist.

2. **Even in d:** evenness of the profile (item 1) gives $w[k,j,-d] = w[k,j,d]$, so only $d \ge 0$ need be stored. (The *source* need not be symmetric in rapidity; the *profile* is.)

For $N_{p_T} = 39$, $N_q = 121$, $N_\eta = 61$, $h_\eta = 0.05$, $L \approx 40$: dense W is $(39\cdot61)\times(121\cdot61) \approx 1.4\times10^8$ entries ($\sim$140 MB), the tensor is $39 \times 121 \times 41 \approx 1.9 \times 10^5$ entries ($\sim$1.5 MB) — a factor $\sim 10^2$, at zero accuracy cost, because it encodes exact symmetries (translation invariance and parity of the profile) rather than a numerical truncation.

Two further exact properties shrink the tensor:

1. **Banded:** the Θ-support limits $|\Delta\eta| \le \operatorname{arccosh}\big[(C+Q)/M\big]$, in practice 1–2 units of rapidity, so $B_d = 0$ for $|d| > L$ with $L = \lceil \Delta\eta_{\max}/h_\eta \rceil$; only $2L+1$ offsets exist.

2. **Even in d:** the kernel weight depends on $\Delta\eta$ only through $\cosh\Delta\eta$, so $w[k,j,d] = w[k,j,-d]$ and only $d \ge 0$ need be stored. (The *source* need not be symmetric; the *profile* is.)

For $N_{p_T} = 39$, $N_q = 121$, $N_\eta = 61$, $h_\eta = 0.05$, $L \approx 40$: dense W is $(39\cdot61)\times(121\cdot61) \approx 1.4\times10^8$ entries ($\sim$140 MB), the tensor is $39\times121\times41 \approx 1.9\times10^5$ entries ($\sim$1.5 MB) — a factor $\sim 10^2$, at zero accuracy cost, because it encodes an exact symmetry rather than a numerical truncation.

6.3 Assembly

The assembly generalizes the scatter of Sec. 5 by one stencil dimension. Fix a Fourier mode n and run the same $(q_T, \Delta\eta)$ quadrature loops as the kernel (in $\Delta\eta$ space — see the remark below); a node at $(q_T, \Delta\eta)$ with total quadrature factor

$$\mathrm{fac} = \mathrm{pref}\cdot v_i^q h_q\, q_T\, h_\eta\, \frac{v_j^\theta}{\sqrt{Q^2 - D^2}}\, 2T_n(Z) \tag{57}$$

(the 2 is the two azimuthal roots only — the $\pm\Delta\eta$ fold is *not* performed here, each sign now lands on different η columns) deposits into the tensor via the product of the q_T stencil and the η-offset stencil:

$$w[k,\, j_0+m-1,\, d_0+m'-1] \mathrel{+}= \text{fac}\; L^q_m(q_T)\; L^\eta_{m'}\big(\Delta\eta/h_\eta\big), \qquad m,m'=1,\ldots,P, \tag{58}$$

where d_0 is the stencil start of the offset $\Delta\eta/h_\eta$ on the integer offset axis. Since the kernel factor is even in $\Delta\eta$, only $\Delta\eta \geq 0$ need be traversed and the result stored for $d \geq 0$; the negative offsets are recovered at application time. No loop over η_p exists anywhere: one sweep per (k,j) column of nodes builds the profile valid for *every* daughter rapidity, so the assembly cost equals that of the boost-invariant matrix times the P-fold extra stencil work — *not* times N_η. This is the payoff of the Toeplitz structure on the assembly side.

Two remarks. First, the Z-space form of Sec. 5.1 does *not* apply here: the η-offset stencil needs the rapidity $\Delta\eta$ itself, so the assembly stays in $\Delta\eta$ space (or would have to pay an arccosh per node to recover $\Delta\eta$ from c — exactly the function the Z form eliminates). Second, the offset variable is interpolated on the *uniform* η grid, so the uniform-grid Lagrange weights apply verbatim; the interpolation error in the η direction is the same $\mathcal{O}(h_\eta^P)$ as in q_T.

6.4 Application

Using the evenness $w[k,j,-d] = w[k,j,d]$, the apply is a folded discrete convolution over the stored $d \geq 0$ half:

$$\mathcal{F}_b\big(p_T^{(k)},\eta_a\big) = \sum_j \Big[\, w[k,j,0]\,\mathcal{F}_a\big(q_T^{(j)},\eta_a\big) + \sum_{d=1}^{L} w[k,j,d]\,\big(\mathcal{F}_a\big(q_T^{(j)},\eta_{a+d}\big) + \mathcal{F}_a\big(q_T^{(j)},\eta_{a-d}\big)\big)\Big]. \tag{59}$$

Organizing the computation per offset makes each term a dense $(N_{p_T} \times N_q)$ matrix acting on an η-shifted copy of the source — $L+1$ small GEMMs over the $(q_T \times \eta)$ source array:

$$F_b \;=\; \sum_{d=0}^{L} B_d\,\big[S^{+d}F_a + (1-\delta_{d0})\,S^{-d}F_a\big], \qquad B_d = w[\,\cdot,\cdot,d\,], \tag{60}$$

with $S^{\pm d}$ the shift along η. Cost: $\mathcal{O}(N_{p_T}N_qN_\eta L)$ multiply–adds ($\approx 2\times10^7 \sim$ milliseconds at N_{p_T}=39, N_q=121, N_η=61, $L\approx$40), organized as BLAS-3 rather than the BLAS-2 of a dense block matrix. For large N_η the Toeplitz structure admits FFT application along η, $\mathcal{O}(N_{p_T}N_qN_\eta \log N_\eta)$ — worthwhile only once $L \gg \log N_\eta$, i.e. for very fine rapidity grids; at the sizes above the direct fold wins.

For azimuthally asymmetric, rapidity-dependent sources the procedure is per mode as before: one tensor w_n per Fourier mode, assembled with $2T_n(Z)$, applied to the mode functions $\mathcal{F}_{a,n}(q_T,\eta_q)$.

6.5 Edges and grid layout

Translation invariance fails only at the ends of the rapidity grid: daughter points within $\Delta\eta_{\max}$ of the boundary require parent values beyond it. Since physical sources vanish rapidly at large $|\eta|$ (e.g. $e^{-m_T\cosh\eta/T}$), the consistent convention is zero-padding — the same "source vanishes off the grid" convention used in q_T — with the η grid extended by $\Delta\eta_{\max} = \max_{q_T}\operatorname{arccosh}[(C+Q)/M]$ (evaluated at the band edge; typically 1–2 units) beyond the rapidity window where results are

needed. Concretely: to output $|\eta_p| \le \eta^{\mathrm{obs}}$, the parent grid must span $|\eta_q| \lesssim \eta^{\mathrm{obs}} + \Delta\eta_{\max}$, and the truncation error inherits the source's decay at the padding edge — for thermal-like sources this is $e^{-m_{aT}(\cosh(\eta^{\mathrm{obs}}+\Delta\eta_{\max})-1)/T}$, negligible already for modest padding.

The memory layout follows the apply: storing w as (k, j)-major slices per offset d (i.e. $L{+}1$ dense $N_{p_T} \times N_q$ matrices B_d) makes each term of (59) a contiguous GEMM; `Float32` storage is appropriate for the same reason as in Sec. 5 (rounding 6×10^{-8}, far below the quadrature setting), halving the $\sim$ 1–3 MB per decay and mode.

7 Summary

case	object	size / cost
η-independent, per mode n	dense $W_n \in \mathbb{R}^{N_{p_T} \times N_q}$	$\sim$40 KB, apply $\sim \mu$s
η-dependent, per mode n	tensor $w[k, j, d]$, $\lvert d\rvert \le L$, $d \ge 0$	$\sim$1 MB, apply $\sim$ms

Decay chains compose by matrix multiplication (and, in the Toeplitz case, by convolution of profiles), so the total feed-down of a hadron list is itself a precomputed linear map.